

Exp: 4 Write the Python Program for Crypt-Arithmetic Problem.

Aim:

To write a python program for Crypt-Arithmetic problem.

Algorithm:

Step 1: Write down the arithmetic problem in full, including any carry-overs.

Step 2: Replace each letter with a unique digit from 0 to 9.

Step 3: Generate all possible combinations of digits for each letter, subject to any constraints ' from the carry-overs and the rules of arithmetic.

Step 4: For each combination of digits, evaluate the arithmetic problem and check if it satisfies the given constraints.

Step 5: If a solution is found, output the values of the letters and the arithmetic problem, and exit the algorithm.

Step 6: If no solution is found, output a message indicating that no solution exists.

Program :

```
from itertools import permutations
```

```
letters = ('S', 'E', 'N', 'D', 'M', 'O', 'R', 'Y')
```

```
for p in permutations('0123456789', 8):
```

```
    d = dict(zip(letters, p))
```

```
    if d['S'] == '0' or d['M'] == '0':
```

```
        continue
```

```
    send = int(d['S'] + d['E'] + d['N'] + d['D'])
```

```
    more = int(d['M'] + d['O'] + d['R'] + d['E'])
```

```
    money = int(d['M'] + d['O'] + d['N'] + d['E'] + d['Y'])
```

```
    if send + more == money:
```

```
print("Solution Found")
print("SEND =", send)
print("MORE =", more)
print("MONEY =", money)
print("Letter Mapping:")
for k, v in sorted(d.items()):
    print(k, "=", v)
break
```

Output:

```
>>>
Solution Found
SEND = 9567
MORE = 1085
MONEY = 10652
Letter Mapping:
D = 7
E = 5
M = 1
N = 6
O = 0
R = 8
S = 9
Y = 2
>>> |
```

Result:

The output has verified.